UNBLIND

Build plan for Izzy

Post-dosing module, the FDA blinding report and the science of the evaluation

Your mission tonight: Turn everything the trial collected into evidence a regulator and a scientist can trust, and make that evidence comparable across trials.

Why this is yours: You designed the UNBLIND correction (the meta-regression and the reasoning behind it), and your physics and computer science background suits the module where the statistics and the scientific honesty matter most: the expectancy-adjusted effect, the bias gate on researcher questions, the standard data format and the evaluation.

At a glance	
Deadline	Working prototype on main by 00:00 tonight. Write-up, pitch and rehearsal tomorrow.
Your teammates	Danny and Callum build the other two modules in parallel. You only meet through the shared contract.
How to read this	Sections 1 to 5 are the same in all three plans: the whole project. Your personal sections follow, then the shared timeline, demo script and glossary.
Golden rules	Contract frozen at 20:00. Own your folder. Every agent works with no API key. Every decision is logged. Every trial-changing action goes to a human. Synthetic data is labelled.

1. The project in one page

UNBLIND is an agentic platform that runs a psychedelic clinical trial with a team of AI agents, from protocol design to the FDA report, so the trial measures and controls the bias that sinks these drugs today.

The problem in one breath

To prove a drug works, you compare a drug group with a placebo group and nobody knows who got what. Psychedelics break this: patients know they tripped, and so do the clinicians scoring them. Every result becomes drug effect plus expectation, stuck together. The FDA's final psychedelic guidance (July 2026) names this functional unblinding as the central threat to validity, and it was central to the 2024 rejection of the MDMA application.

Why it stays unsolved

Nobody owns blinding as a job. Most trials never measure it, the ones that do measure it differently, and problems are only discovered at the end, after the money is spent. Trials are also slow to fill, and screening volunteers is manual.

What we are building

A master agent coordinates three modules, one per phase of a trial. Each module is a small team of agents that does real work every simulated day and brings decisions to a named human.

Phase	What the agents do	Owner
Pre-dosing	Co-design the protocol (risk, fixes, drafted blinding section, re-check on every edit), write ethics-ready recruitment ads, pre-screen applicants automatically, build the standard measurement plan and schedule	Danny
Dosing (acute)	Run post-session check-ins (guess, confidence, expectancy), structure monitor debriefs, monitor blinding per site and per rater through a sealed vault, detect drift, propose actions, chase missing data, scan new literature weekly	Callum
Post-dosing	Standard follow-up collection, a researcher question portal with a bias gate, a harmonised data export, and the FDA blinding report with an expectancy-adjusted effect	Izzy
Master agent	Runs the clock, routes messages between agents, keeps the trial memory, queues human decisions, shows mission control	Danny

What the judges will see

- A judge pastes or edits a draft protocol. The pre-dosing agents score its blinding risk, fix it, and screen 260 synthetic applicants down to a clinician-approved shortlist.
- Press play. Simulated weeks pass in about two minutes: sessions, check-ins, debriefs and a live blinding chart per site.
- Halfway through, raters at one site start guessing right far too often. The dosing agents spot it, explain it and propose moving those raters to remote assessment. The judge clicks approve.
- The judge can play a patient and answer a check-in.
- At close-out the post-dosing agents produce a standard FDA blinding report: every measurement, alert and human decision, plus the reported and expectancy-adjusted effect.

Everything simulated is SYNTHETIC and labelled as such, which the hackathon rules allow. No real patient data, ever.

2. The bottlenecks, quantified and sense-checked

The brief says every claim must name a bottleneck, quantify it, and show exactly how we move it. These are the numbers we can stand behind tonight. Anything not listed here needs a source before it goes in the pitch.

Bottleneck	Number (with source)	How UNBLIND moves it	Metric we report
Blinding is rarely measured	Only 29.5% (33 of 112) randomised psychedelic trials assessed blinding (systematic review, JAMA Psychiatry 2026)	Every participant and rater gets the same scheduled guess check-in by default	Share of scheduled blinding check-ins completed
When measured, blinding fails	Psilocybin, LSD and ayahuasca studies frequently reported failure above 90%; inert-placebo MDMA trials above 85% (same review)	Risk predicted before the trial; drift caught per site and per rater within days	Days from drift onset to alert
Regulators now demand it	FDA final guidance, July 2026: blinding questionnaires, expectancy measures before randomisation and at end of treatment, central blinded raters, alternative controls	Protocol checked against each item; the evidence collected into one report	Guidance items satisfied, before and after
Sessions are labour-heavy	Two monitors per session and a physician within 15 minutes (FDA guidance)	Debriefs spoken or typed, structured automatically into adverse events and unblinding cues	Debriefs structured automatically
Effects are small	COMP005: -3.6 MADRS vs placebo (95% CI -5.7 to -1.5). COMP006: -3.8 vs 1 mg psilocybin (95% CI -5.8 to -1.8) (Compass press releases)	Close-out report shows reported and expectancy-adjusted effect side by side	Reported vs adjusted effect, with interval
Blinding data is not comparable	Of 18 real trials our research agent examined, only 4 report clean per-arm guess data; timing ranged from dosing day to day 365 (our candidate gold set, unverified)	One standard instrument set, UNBLIND-BLIND-1, at fixed timepoints, exported in one schema	Trials exporting in the standard schema
Recruitment and screening are slow and manual	NEEDS A SOURCE. Do not quote a number until Callum or Izzy finds one	Agent screens every applicant in seconds; clinicians review only a shortlist and the flagged cases	Share screened automatically; clinician reviews per enrolled participant; staff hours saved (state the per-applicant minutes as an assumption)

Sense checks: what a scientist judge will push on

- **Bigger trials do not reduce bias.** More patients shrink random error, not the systematic error from knowing your group. A 1,000-person trial with broken blinding is just precisely wrong. Pitch recruitment as faster and fairer, not as less biased. Also, phase 3 psychedelic trials are already large (COMP006 dosed 581), so say 'most academic phase 2 trials are small'.
- **Pre-screening cannot remove humans entirely.** A clinician legally confirms final eligibility, especially for psychosis history, heart conditions and medication. The agent automates the funnel so humans review only a shortlist. That is the honest efficiency claim.
- **Prior psychedelic use: balance it, do not exclude it.** Experienced users recognise the drug faster, but excluding them shrinks the pool. Stratified randomisation spreads them evenly across arms.

- **Targeted ads using health or drug-use interests are risky** under UK GDPR (special category data) and are restricted by ad platforms. We park targeting and build compliant, ethics-ready ads instead.
- **Researcher questions can themselves break blinding.** Asking about 'mystical experiences' before the primary result is locked tells patients what they got. Every question passes a bias gate and is released only after the primary lock.
- **Reverse causation.** Patients who improve may guess 'drug' because they improved, so correcting for guesses can remove real effect. The adjusted effect is a conservative bound, not the truth. Our synthetic test shows the size of this failure (about 0.15 g over-correction).
- **Facts to say correctly:** the 3.8 is COMP006 against 1 mg psilocybin, not placebo. 'Above 90% failure' applies to many psilocybin, LSD and ayahuasca studies, not to all trials. Compass has not been 'cleared': rolling review is a submission route, not a verdict.
- **Prior art:** Szigeti and colleagues published an 'equal unblinding' meta-analysis in 2026. Our novelty is the agentic, living, design-time platform, not the idea that blinding matters.

3. How the platform works

The shared foundations (the only things modules share)

Piece	What it is	Why it matters
TrialRecord	One JSON document per trial holding the protocol, applicants, schedule, check-ins, metrics, alerts, approvals and events. Each module writes only its own sections.	Lets three people build in parallel without touching each other's code
MessageBus	Agents post messages (for example 'rater_drift'). The master agent routes them to whoever needs to know.	Agents cooperate without importing each other
ApprovalQueue	Any action that changes the trial becomes a request. A named human approves or rejects. Everything is logged.	The brief's 'human signs off key decisions', made concrete
Vault	The only holder of who got the drug. Agents receive group-level statistics only, with groups under 5 suppressed. Unsealing needs a named person and is logged.	Security: the AI cannot leak what it never sees
World	A SYNTHETIC simulator of applicants, patients, sites and raters, with planted events (rater drift at Site B from day 28, missing data at Site C, exclusions hidden in free text).	Lets us demo months in minutes, and evaluate agents against a known truth

The master agent

- Advances the simulated clock one day at a time and sets the phase: design, dosing, follow-up, close-out.
- Delivers messages and decides who hears about what. For example, rater drift goes to pre-dosing (draft a protocol amendment) and to post-dosing (record it in the FDA report). With a Gemini key it reads the message and chooses recipients; without one it uses a routing table. Either way the choice is logged.
- Isolates failures: if one module crashes, the others keep running and the error appears in the timeline.

- Writes a weekly summary and powers mission control: phase bar, timeline, approval inbox, agent message map, KPIs.

The simulated timeline

Day	Phase	What happens	Module
Setup	Design and recruitment	Protocol parsed and reviewed, fixes proposed, ads written, 260 applicants screened, clinician approves shortlist, stratified randomisation in the vault, measurement plan and schedule built	Pre-dosing
1 to 20	Dosing	Staggered dosing sessions; monitor debrief the same day; participant check-in the next day	Dosing
Each participant's weeks 1, 3, 6	Dosing	Rater visit: MADRS score plus rater guess; participant check-in	Dosing
Daily	Dosing	Blinding metrics per site and rater, drift detection, missing-data chasing	Dosing
Weekly	All	Literature scan with human approval; master weekly summary	Dosing and master
After last week-6 visit	Primary lock	Dosing tells the master the primary endpoint is locked	Dosing to master
Weeks 8 and 12	Follow-up	Standard follow-up collection; researcher questions released through the bias gate	Post-dosing
Day 90	Close-out	FDA blinding report and harmonised export, signed off by a human	Post-dosing

What already exists and gets reused

- **The UNBLIND research pipeline** (already in the repo): eight agents that read any trial and extract its facts with verified quotes, a deterministic quote auditor, the meta-regression, the design advisor, a Streamlit app, an evaluation suite and 18 candidate real trials. In the platform it becomes the weekly literature agent (dosing) and the risk engine (pre-dosing and post-dosing).
- **A starter scaffold** drafted by Claude in its own workspace: core.py (record, bus, approvals, vault), world.py (simulator), master/orchestrator.py and MODULE_CONTRACT.md. It is not in the repo yet. Danny decides at kick-off whether to pull it in; Claude can sync it on request.

4. How this scores against the judging criteria

Criterion	What a top score needs	Where UNBLIND delivers it	Owner
Uniqueness	Beyond a literature chatbot or stock model	Agents that live inside a trial for months; a vault the AI cannot read; a bias gate on researcher questions	All
Real-world need	One specific bottleneck, named user, plausible next step	Unmeasured, failing blinding. User: the clinical development lead running a psychedelic trial. Next step: pilot the standard kit with one contract research organisation	Izzy (narrative)
Science, data and evaluation	Tested against a baseline, data referenced and licence-checked, uncertainty stated	Drift detection vs end-of-trial-only checking; screening vs synthetic ground truth; adjusted vs naive effect on planted-truth simulations; sources and licences listed	Callum and Izzy

Criterion	What a top score needs	Where UNBLIND delivers it	Owner
Agent design and live demo	Real multi-step agent; demo handles a judge's new input	Judge edits a protocol, approves actions, plays a patient, submits a researcher question	Danny (shell), everyone (moments)
Security and responsible design	Named risks with mitigations; human signs off key decisions	Vault, approval queue, quote audit, bias gate, consent checks, GDPR notes, synthetic labelling	All

5. Who does what

Person	Owns	Mission
Danny	Pre-dosing module, master agent and mission control (platform lead)	Make the trial start right, and make the whole platform run as one product.
Callum	Dosing (acute) module: the agents that live inside the trial	Catch blinding failure while the trial can still fix it.
Izzy	Post-dosing module, the FDA blinding report and the science of the evaluation	Turn everything the trial collected into evidence a regulator and a scientist can trust, and make that evidence comparable across trials.

Izzy, the next sections are yours. Everything below is written for your module. Your teammates have the same shared sections and their own versions of these.

6. Your module: post-dosing

The bottlenecks you attack

- **Blinding data is not comparable.** Of 18 real trials examined, only 4 report clean per-arm guess data, at timepoints from dosing day to day 365, from different respondents. Your harmonised export (UNBLIND-BLIND-1) makes every trial on the platform comparable by design.
- **Effects are small enough that bias matters** (COMP005 -3.6 and COMP006 -3.8 MADRS points). Your close-out report shows the reported effect next to an expectancy-adjusted range, with the method and its limits.
- **Follow-up research can break blinding or plant expectations.** Your bias gate makes outside questions safe and releases them only after the primary lock.
- **Sense check to own:** reverse causation. The adjusted effect is a conservative bound. You present it that way, with the simulation showing the size of the failure.

Your agents

Agent	What it decides (the agentic part)	Output	Human gate	No-key fallback
Follow-up agent	Runs weeks 8 and 12: MADRS, end-of-treatment guess and expectancy, durability; chases non-responders	rec['followups']	None	Deterministic
Bias gate agent	Reviews each researcher question for (1) unblinding cues (visuals, intensity, 'the drug'), (2) leading or expectancy wording, (3) identifying or personal data, (4) safety content needing a clinician, (5) timing and consent. Decides approve, rewrite (and proposes neutral wording) or reject, with reasons.	rec['research_questions']	New questions need an ethics approval request before release	Rules and word lists; Gemini rewrite if a key is set

Agent	What it decides (the agentic part)	Output	Human gate	No-key fallback
Participant interview agent	Asks approved questions only to participants who consented to data sharing, after the primary lock; summarises answers into themes without identifiers	Answers and themes	None beyond the gate	Keyword themes
Harmoniser	Builds the UNBLIND-BLIND-1 export: pseudonymised IDs (salted hash), consent filter, data dictionary; notes the future CDISC mapping	rec['exports'] and files	Export needs approval	Deterministic
Close-out report agent	After the primary lock: reported effect from vault.unblinded_effect (aggregate), blinding indices over time per site and rater, every alert and human decision, the FDA checklist with evidence, and the expectancy-adjusted range with method and limitations	rec['report'] (markdown and JSON)	A named person signs off the report	Deterministic

The expectancy-adjusted effect (your science centrepiece)

- **Cross-trial method:** apply the existing calibration (unblind.model) with this trial's observed u. Say clearly when the calibration is synthetic.
- **Within-trial method (new, and stronger):** because every participant now has a guess recorded before the week-6 outcome, the vault can return aggregate outcomes by arm and guess. Compare drug vs control within participants who guessed the same way. Ask Danny for Vault.effect_by_guess() at 20:00 (aggregates only, cells under 5 suppressed).
- **Validate against the world's truth:** the simulator plants a true effect of 4 MADRS points, plus expectancy (3) and rater bias (2). Report the naive effect, the adjusted effect and whether the interval contains 4, across 20 seeds. This is the honest 'tested against a baseline' result.
- **State limits on the report itself:** reverse causation, confounding, synthetic calibration.

The standard: UNBLIND-BLIND-1

Field	Meaning
pid	Pseudonymised participant ID (salted hash)
site, stratum	Site and naive or experienced
timepoint_day, visit_week	When the measure was taken, relative to dosing
respondent	participant, rater or monitor
guess, confidence, expectancy	drug, control or don't know; 0 to 100; 0 to 10
madrs	Outcome score where applicable
consent_share	Whether the row may be exported
source_event_id	Link back to the audit trail

7. Your files and interfaces

You own	You read	You send	You receive
unblind/platform/postdose/*, eval/platform/*, tests/platform/test_postdose.py	rec['enrolled'], rec['checkins'], rec['outcomes'], rec['blinding_metrics'], rec['alerts'], rec['approvals'], rec['design_review'], schedule entries with owner 'postdose'	question_rejected (log only)	primary_locked (start follow-ups and open the portal), rater_drift, adverse_event, missing_data (record in the report)

8. Your task list

#	Time	Task	Priority
1	19:45	Read the contract; request Vault.effect_by_guess() from Danny	Must
2	20:00	Follow-up agent on schedule entries owned by postdose	Must
3	20:20	Bias gate agent with a test set of 15 good and bad researcher questions	Must
4	20:50	Close-out report: reported effect, blinding timeline, alerts and decisions, checklist, adjusted range, limitations	Must
5	21:30	Checkpoint: full Master run to day 90	Must
6	21:45	Post-dosing page: question portal (judge submits a question), gate decisions with reasons, follow-up status, report preview with sign-off	Must
7	22:30	Evaluation over 20 seeds: naive vs adjusted vs truth (4 points); write eval/platform/results.md	Must
8	22:50	Harmonised export with data dictionary and consent filter	Should
9	23:00	Integration fixes	Must
10	later	Participant interview agent with theme summaries; Gemini rewrites	Could

9. Research tasks

- Verify 6 rows of data/gold_candidates.csv against their sources (quote and number); mark each with your name. This feeds the existing pipeline's extraction evaluation.
- Write the 'what is new vs Szigeti et al. 2026' paragraph for the write-up: two sentences, honest.

10. Tests, evaluation and judge questions

- **Tests:** no question is released before the primary lock; leading or unblinding questions are rewritten or rejected; non-consenting participants are never asked or exported; the export has no names or phone numbers; the report cannot be finalised without a named sign-off.
- **Headline numbers:** naive vs adjusted effect vs planted truth across seeds (bias and coverage); bias-gate accuracy on the labelled question set; export completeness.
- **'Is the adjusted number the true effect?'** No, it is a bound under stated assumptions. Here is how wrong it gets when patients guess from improvement.
- **'Why should other trials use your format?'** Because the FDA asks for these measures without saying how; a shared format makes every trial comparable and improves every prediction.

- **'Has this been done?'** Equal-unblinding meta-analyses exist (Szigeti et al. 2026). Nobody runs the measurement as a living, standard, agent-operated process inside trials.

11. Tomorrow

- Lead the one-page write-up.
- Open and close the science part of the pitch (problem, and the post-dosing evidence).
- Own the limitations slide.

12. How we work in parallel tonight

- **Contract freeze at 20:00.** After that, nobody changes the shared shapes in MODULE_CONTRACT.md without telling the other two in the group chat.
- **Own your folder.** Danny: unblind/platform/predose/, master/, core.py, world.py and app/platform_app.py. Callum: unblind/platform/dosing/. Izzy: unblind/platform/postdose/ and eval/platform/. Nobody edits another person's folder.
- **Git.** One branch each (danny/predose, callum/dosing, izzy/postdose). Pull main at every checkpoint. Danny merges.
- **Every module works with no Gemini key.** Deterministic fallback first, Gemini second. The demo must survive bad Wi-Fi.
- **Every agent explains itself.** rec.event(agent, text, level, module) for each decision. Decisions are what make it look and behave like an agent.
- **Human gate.** Anything that changes the trial goes through the ApprovalQueue.
- **Vault only.** Never compute anything from allocation outside the vault.
- **Label synthetic data** on every page and in every chart.
- **Copy:** UK English, plain language, no em dashes.

Tonight's timeline

Time	Everyone	Checkpoint
19:45 to 20:00	Read this plan. Agree the contract. Danny sets up branches and the scaffold.	Contract frozen
20:00 to 21:30	Build block 1: agents with deterministic fallbacks, plus offline tests	Each module runs inside the master with the others present or missing
21:30 to 21:45	Checkpoint call: run Master end to end on main	Known bugs listed
21:45 to 23:00	Build block 2: module page, KPIs, Gemini enhancements, your demo moment	Pages render in mission control
23:00 to 23:30	Integration: Danny runs the full demo path twice; others fix their bugs	Clean run from setup to close-out
23:30 to 00:00	Freeze. One rehearsal of the demo path. Push and tag v1.	Working prototype on main

Tomorrow

Task	Lead	Support
One-page write-up (problem, who it helps, data, evaluation, limits and safeguards)	Izzy	Callum (evaluation numbers), Danny (architecture)
Pitch deck and 4-minute script	Danny	Izzy (science slides), Callum (evaluation slide)
Judge Q&A sheet (sense checks above, with answers)	Callum	Izzy
Demo rehearsal x3, including a judge's unexpected input	Danny	All
README and repo tidy	Danny	All

13. Demo script (4 minutes + 2 minutes for judges)

Time	Speaker	On screen
0:00 to 0:30	Izzy	The problem: patients know they tripped. 29.5% of trials measure blinding; many fail above 90%. FDA July 2026 names it the central threat.
0:30 to 1:30	Danny	Pre-dosing: paste a draft protocol. Risk score, ranked fixes, FDA checklist, drafted blinding section. Approve a fix and watch it re-check. 260 applicants screened to a shortlist; flagged free-text disclosures shown with quotes.
1:30 to 2:40	Callum	Press play. Live blinding chart per site. Site B's raters drift; the agent explains and proposes remote raters; approve; drift stops. Show a debrief turned into structured events.
2:40 to 3:30	Izzy	Post-dosing: researcher asks a leading question; the bias gate rewrites or blocks it. Close-out FDA blinding report: reported vs expectancy-adjusted effect, with every decision traceable.
3:30 to 4:00	Danny	What it means: one standard, every trial comparable, a dataset that improves every new trial. Next step: pilot with one CRO.
4:00 to 6:00	Judges	They edit the protocol, play a patient, or submit a question. Everything is live.

14. Glossary

Term	Plain meaning
Blinding	Keeping secret who got the real drug
Functional unblinding	The drug's effects give the secret away
Expectancy	Feeling or scoring better because you believe you got the drug
Guess accuracy, u	Share of correct 'which group?' guesses. u = 2 x accuracy - 1, so 0 is a coin flip and 1 means everyone knew
Bang blinding index	Standard blinding score per arm, from -1 to 1, where 0 is ideal
Central blinded rater	The person scoring the patient is remote, never sees the session, and does not know the arm
Stratified randomisation	Randomising separately within groups (for example, experienced vs naive), so both arms get the same mix
MADRS	A clinician-scored depression scale, 0 to 60, higher is worse
Primary endpoint and lock	The one outcome the trial promised to test; locking it means no more changes before analysis
Hedges' g	Effect size in standard-deviation units, comparable across scales
Meta-regression	A line fitted across many trials, one dot per trial
CRO	Contract research organisation: companies that run trials for drug makers
Synthetic data	Simulated data, labelled as such, never presented as real-world performance